Data and AI Internship Program ADM Capability Center, Bangalore

CASE STUDY SOLUTION

1. GenAI Assistant for Business Knowledge
2. Commodity Pricing & Freight Optimization
3. Data Quality & Pipeline Troubleshooting

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Three problems, one evidence-first approach

The analysis is scoped to the supplied fictional documents and datasets.

GENAI

Grounded business knowledge

Approved documents → retrieval → evidence-backed answer/refusal

PRICING

Delivered-cost optimization

Reconcile formulas → compare product × destination → isolate drivers

DATA QUALITY

Pipeline troubleshooting

Profile records → reconcile revenue → trace pipeline → control publication

METHOD 01 **Inspect** → 02 **Calculate** → 03 **Validate** → 04 **Interpret** → 05 **Recommend**

Scope note: all case datasets are fictional training data; findings should not be generalized to real ADM operations.

Source: ADM case materials • fictional assessment data

Grounded Q&A: answer from evidence, not memory

The case references five sample questions, but those five questions are not included in the supplied materials.

ILLUSTRATIVE GROUNDED TEST QUESTIONS

Business question	Grounded answer	Evidence
What is the lead time to Chennai?	**3 business days**	Shipping Guidelines
Who approves a discount above 2%?	**Regional approval**	Pricing Policy
Can the assistant provide a final customer quote?	**No — escalate to the authorized commercial process.**	Pricing Policy
What if approved documents do not support an answer?	**State it is unavailable; direct to the business owner.**	FAQ & Governance

Design rule **If evidence is missing or conflicting: do not guess → disclose the gap → escalate.**

SIMPLE RETRIEVAL FLOW

1. **Question**
2. **Retrieve approved passages**
3. **Check policy / metadata**
4. **Answer + evidence**
5. **Refuse / escalate if unsupported**

Source: Product Catalog • Shipping Guidelines • Pricing Policy • FAQ & Governance

Controls that make the assistant trustworthy

A narrow RAG design fits a small, policy-driven approved knowledge base.

PROPOSED SYSTEM PROMPT

"Answer only from approved retrieved documents. Do not invent or fill gaps with general knowledge. Cite the supporting document. If evidence is insufficient, say so and escalate. Never provide a final customer quote."

RISK → CONTROL

Hallucination	Evidence-only generation + refusal
Outdated policy	Version / effective-date metadata
Unauthorized access	Document-level access control
Pricing commitment	No final quote + human approval
Sensitive data	Redaction + output checks
Citation mismatch	Citation-to-passage tests

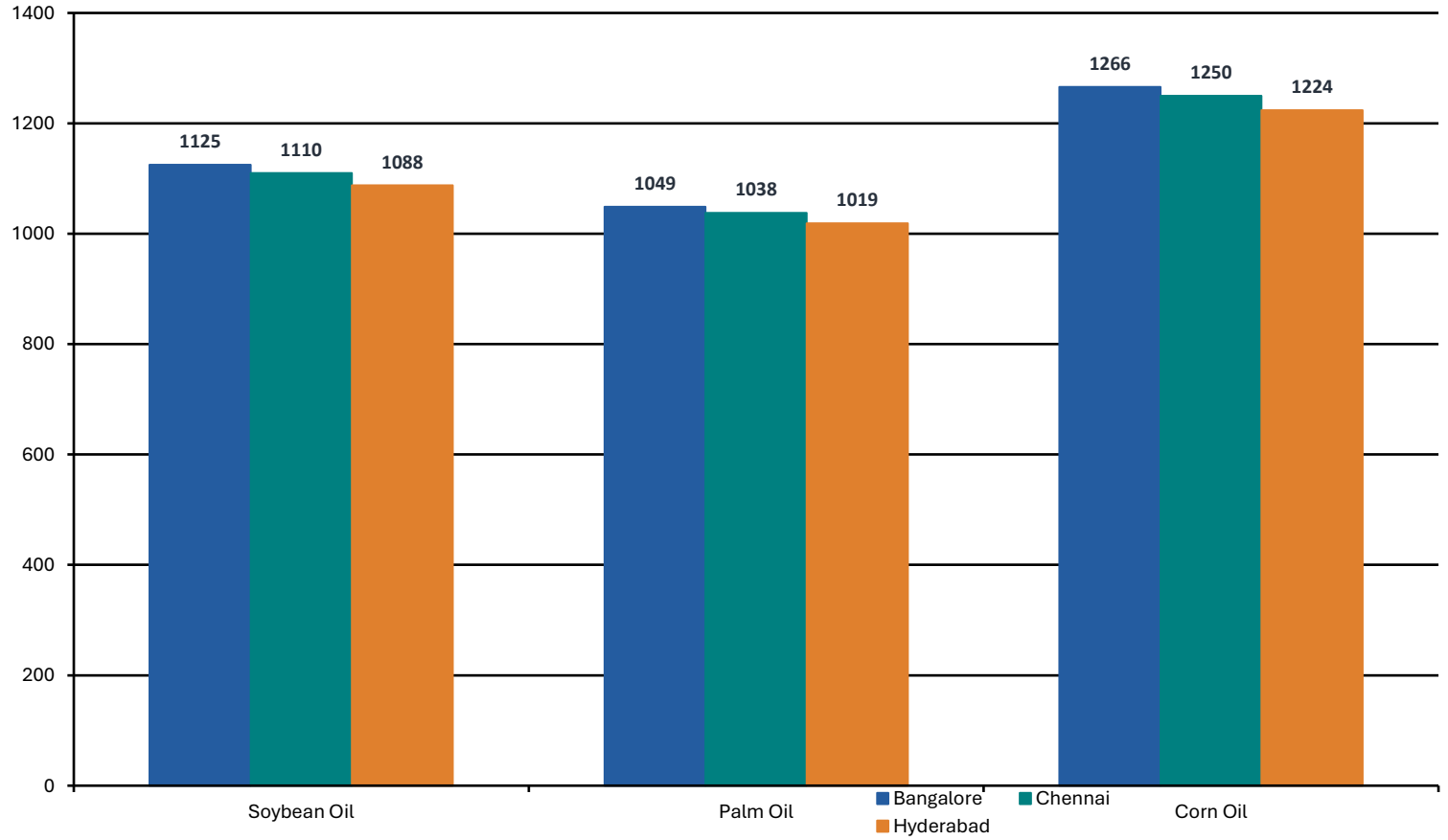
EVALUATION PLAN

T1	**Supported fact**	Correct answer + citation
T2	**Unsupported question**	Refusal / escalation
T3	**Conflicting versions**	Use current approved version
T4	**Pricing request**	No final quote
T5	**Access-restricted doc**	No unauthorized retrieval
T6	**Citation check**	Citation matches evidence

Source: FAQ & Governance • Pricing Policy • Case Instructions

Delivered cost is lowest in Hyderabad across all three products

Average delivered cost per MT from the validated Clean_Data sheet.



WHAT THE DATA SAYS

3 / 3

products: Hyderabad is lowest

₹1,016

lowest observed order cost / MT

18.41M

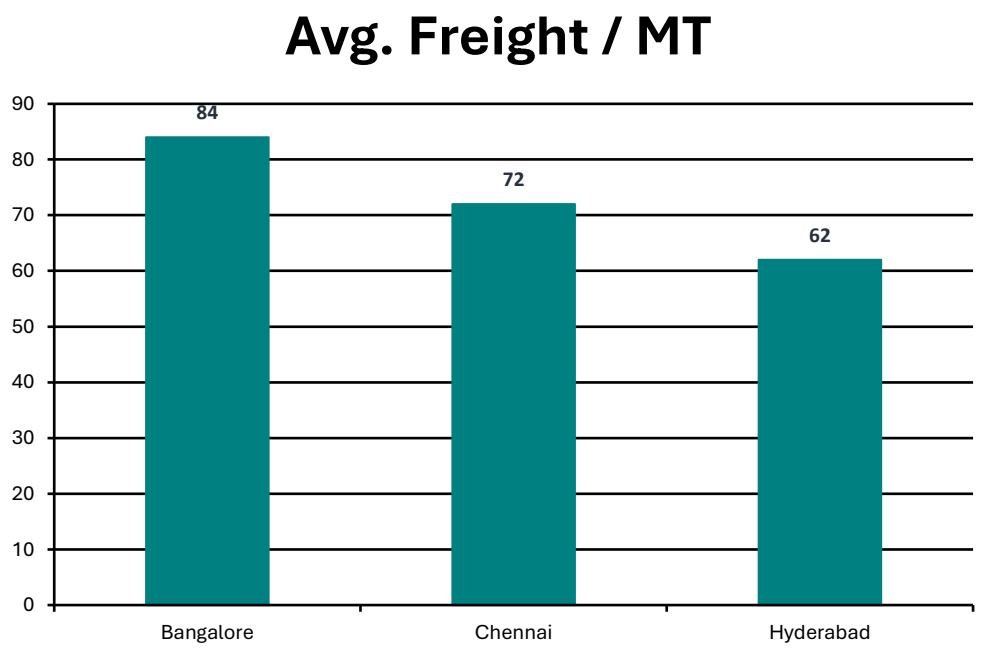
total value of 15 sample orders

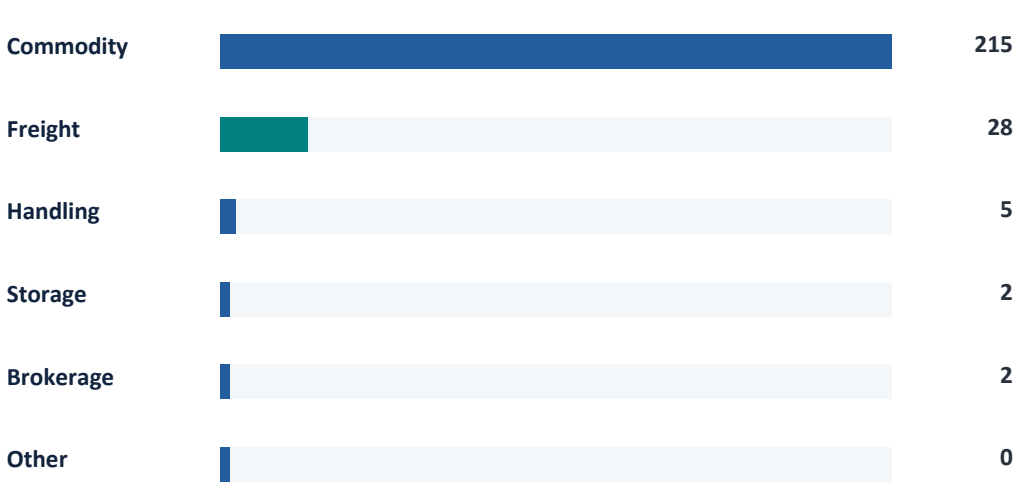
Sample finding—not a universal ADM market claim.

Source: Clean_Data • Data Dictionary • Business Background

Freight is the main destination-sensitive cost lever

Commodity cost is largest overall, but freight explains most of the destination gap.





BUSINESS RESPONSE

Monitor and negotiate freight by destination → use delivered cost as the baseline → validate lead time, capacity and customer commitments before changing lanes.

The pricing practice sheet contains four distinct quality issues

Each issue is traceable to a specific record and has a direct analytical impact.

Issue	Type	Evidence	Business impact
Missing quantity	Completeness	Order 8: Quantity blank; Total Order Value populated.	Volume/value cannot be independently verified.
Missing freight	Completeness	Order 11: Freight blank; delivered cost still populated.	Component-level recalculation becomes unreliable.
Destination whitespace	Consistency	Order 5: “Chennai ” differs from “Chennai”.	Exact-match grouping can split Chennai into two buckets.
Duplicate Order_ID	Uniqueness	Order 3 appears twice identically.	Aggregations can double-count the order.

CONTROL **Profile completeness + uniqueness + standardized dimensions before any product × destination aggregation.**

Source: Data_Quality_Practice • Data Dictionary

The arithmetic reconciles; the data does not fully explain itself

Expected Revenue = Quantity_MT × Unit_Price. Complete records reconcile exactly.

100%

complete records reconcile exactly

2

records cannot be calculated

1

negative-quantity rule violation

1

duplicate order (1010)

RECORD-LEVEL CHECK

Order	Reported	Expected	Result	Comment
1001	117,600	117,600	✓ 0	Valid
1004	118,200	—	N/A	Quantity missing
1006	−22,200	−22,200	✓ 0	Quantity invalid
1007	122,500	—	N/A	Unit price missing
1010	107,800	107,800	✓ 0	Duplicate row
1011	123,525	123,525	✓ 0	Valid

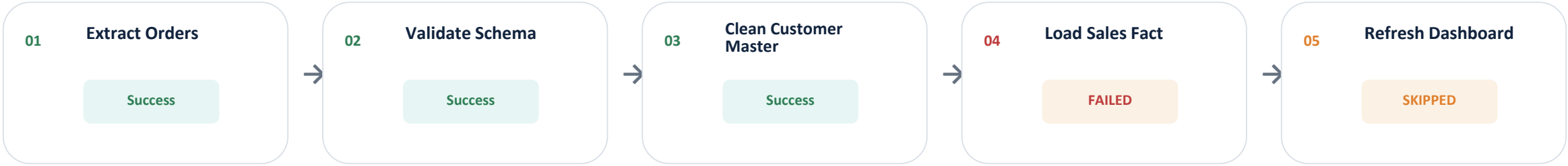
INTERPRETATION

- No arithmetic/pricing formula error is evidenced where Qty and Price are present.
- 1004 and 1007 make reported revenue unverifiable at record level.
- 1006 violates the stated Qty > 0 business rule.
- 1010 can inflate revenue if duplicates are not removed.
- The specific dashboard decline cannot be attributed to one issue from the supplied data alone.

Source: Sales_Data • Pipeline Background • Data Dictionary

The dashboard was blocked by a failed fact-load step

The pipeline log gives the immediate failure cause; the deeper reason for the timeout is not specified.



WHAT FAILED

Load Sales Fact

Database connection timeout

Dashboard refresh was skipped because the upstream load failed.

AUTOMATED QUALITY GATES

Schema & completeness	Required fields; valid types; null thresholds
Business rules	Qty > 0; Unit Price > 0; unique Order_ID
Revenue reconciliation	Qty × Price = Reported Revenue where fields exist

Deep root cause of the timeout: not specified in the supplied case materials.

Publication rule: fail safely — do not refresh on a partial / failed load.

Source: Pipeline_Run_Log • Pipeline Background • Sales_Data

Three actions to turn analysis into repeatable decisions

Keep the controls simple, measurable and tied to the evidence in this case.

01

TRUSTED GENAI

Ground every answer in approved retrieved evidence.
Refuse unsupported questions; cite sources; require human review for pricing exceptions.

02

COST INTELLIGENCE

Make delivered cost the routing baseline.
Monitor destination freight and validate lead time, capacity and commercial commitments.

03

DATA RELIABILITY

Gate publication on completeness, validity, uniqueness and revenue checks.
Alert on critical pipeline failures; publish only trusted data.

BOTTOM LINE

Trusted data → explainable AI → better business decisions

All conclusions are based on the supplied fictional assessment materials and should be validated against current business data before production decisions.